

BOA JANG

Seoul National University, Interdisciplinary Program in Bioengineering

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BIOGRAPHY	I develop robust and generalizable deep learning methods for medical imaging, in close collaboration with clinicians. My research interests include domain adaptation and generalization, multimodal medical AI, generative modeling for clinical imaging, and imaging biomarkers. My goal is to build medical AI that remains reliable across devices, hospitals, and populations — and creates real impact in clinical care.	
EDUCATION	Seoul National University	Seoul, Korea
	<i>Ph.D. Candidate in Bioengineering</i>	Mar. 2023 – Present
	Peking University	Beijing, China
	<i>B.S. in Psychological and Cognitive Sciences</i>	Sep. 2016 – Jul. 2020
RESEARCH EXPERIENCE	Seoul National University Hospital (SNUH)	Seoul, Korea
	<i>Medical AI Researcher</i>	Nov. 2021 – Feb. 2023
	Korea Institute of Science and Technology (KIST)	Seoul, Korea
	<i>Research Intern, Brain Science Institute</i>	Aug. 2020 – May 2021
	Peking University	Beijing, China
JOURNAL PUBLICATIONS	<i>Undergraduate Research Assistant</i>	Mar. 2019 – Jun. 2020
	[j.8] J.R. Song [†] , B. Jang [†] , et al. (2026). Segmentation-Guided Denoising Diffusion Probabilistic Models for Retinal OCT Speckle Noise Reduction with Enhanced Clinical Diagnostic Accuracy. <i>Scientific Reports</i> (Accepted).	
	[j.7] B. Jang , Y. Ahn, et al. (2026). Clinically Grounded Retinal Representation Learning from Minimal Supervision. <i>Scientific Reports</i> .	
	[j.6] B. Jang , Y. Shin, G. Lee, and Y.-G. Kim (2026). From Static Diagnosis to Dynamic Guidance: Evolution of Artificial Intelligence in Pediatric Neuroimaging. <i>Journal of Korean Neurosurgical Society</i> .	
	[j.5] J.H. Lee, S.Y. Lee, B. Jang , et al. (2026). OCT Biomarkers as Predictors of Treatment Interval in Neovascular Age-Related Macular Degeneration Treated with Intravitreal Aflibercept Using a Treat-and-Extend Regimen. <i>Scientific Reports</i> .	
	[j.4] B. Jang [†] , C.H. Lee [†] , et al. (2026). Artificial Intelligence-Based Prediction of First Recurrence in Neovascular Age-Related Macular Degeneration with Validation by 19 Experts. <i>Scientific Reports</i> .	
	[j.3] D.Y. Kim, R. Do, . . . , B. Jang , et al. (2025). Automated AI-Based Identification of Autism Spectrum Disorder from Home Videos. <i>npj Digital Medicine</i> .	
	[j.2] N.Y. Kim [†] , B. Jang [†] , et al. (2023). Differential Diagnosis of Pleural Effusion Using Machine Learning. <i>Annals of the American Thoracic Society</i> .	
	[j.1] B. Jang [†] , S.Y. Lee [†] , et al. (2023). Preliminary Analysis of Predicting the First Recurrence in Patients with Neovascular Age-Related Macular Degeneration Using Deep Learning. <i>BMC Ophthalmology</i> .	
	[†] These authors contributed equally.	
PRESENTATIONS	[o.7] B. Jang , G. Lee, J. Choi, and Y.-G. Kim. “Structure-Preserving Test-Time Adaptation for Cross-Modality Retinal Vessel Segmentation.” <i>2026 Spring Conference of The Korean Society of Medical Informatics (KoSMI)</i> , Cheongju, Korea.	
	[po.3] B. Jang , K.H. Bae, and Y.-G. Kim. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” <i>2025 Annual Conference of the Korea Society of Artificial Intelligence in Medicine (KoSAIM)</i> , Korea.	
	[o.6] B. Jang et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” <i>MICCAI 2025 AMAI Workshop</i> , Daejeon, Korea. Best Abstract Award – Honorable Mention.	

UNDER REVIEW	[o.5] B. Jang et al. “Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability.” <i>2025 Spring Conference of The Korean Society of Medical Informatics (KoSMI)</i> , Busan, Korea. Best Paper Award .
	[o.4] B. Jang et al. “Self-Supervised Learning for Mesangial Proliferation Prediction in Digital Pathology.” <i>The 45th Annual Meeting of the Korean Society of Nephrology (KSN 2025)</i> , Seoul, Korea.
	[o.3] B. Jang and Y.-G. Kim. “Foundation Models in Healthcare: Development, Necessity and Implementation Strategies.” <i>2025 HIMSS Global Health Conference & Exhibition (HIMSS 2025)</i> , Las Vegas, NV, USA.
	[o.2] B. Jang et al. “The Impact of AI-Assistance on Predicting Recurrence in Neovascular Age-Related Macular Degeneration for Ophthalmologists.” <i>2023 Fall Conference of The Korean Society of Medical Informatics (KoSMI)</i> , Bundang, Korea. Best Paper Award .
	[po.2] B. Jang et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” <i>10th Workshop on Ophthalmic Medical Image Analysis (OMIA-X), 2023 in International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)</i> , Vancouver, Canada.
	[po.1] B. Jang , Y.-G. Kim, and E.K. Lee. “Predicting the First Recurrence in Patients with Neovascular AMD Using Deep Learning.” <i>2023 American Society of Retina Specialists (ASRS)</i> , Seattle, WA, USA. Poster Award Winner .
	[o.1] B. Jang et al. “Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images.” <i>2023 Spring Conference of The Korean Society of Medical Informatics (KoSMI)</i> , Daegu, Korea. Best Paper Award .
	[m.5] Skeleton-Guided Progressive Test-Time Adaptation for Thin Curvilinear Structures. <i>AAAI2027</i> (under review).
	[m.4] Retinal Age Gap as a Biomarker of Biological Aging Reflected by Socioeconomic Status and Lifestyle Behaviors. <i>GeroScience</i> (in revision).
	[m.3] Prediction of Postoperative Intracerebral Hemorrhage Following Brain Biopsy Using Integrated Statistical and Machine Learning Analyses. <i>Scientific Reports</i> (in revision).
AWARDS	[m.2] Integrating RNFL and Fundus Photographs for Differentiation of Glaucomatous and Non-Glaucomatous Optic Neuropathy Using Deep Learning. <i>Scientific Reports</i> (in revision).
	[m.1] Age- and Sex-Specific Autism Screening Using Machine Learning Reduction and Conversational Agent Integration. <i>PLOS Digital Health</i> (submitted).
	[a.5] AI Seoul Tech Graduate Scholarship, Seoul Future Talent Foundation Nov. 2025
	[a.4] Hana Bank President’s Award, 2024 Digital Wellness Competition Nov. 2024
	[a.3] 2nd Place, Healthcare AI Learning Platform Challenge 2023, Themes 4 & 6 Dec. 2023
	[a.2] 3rd Place, Digital Pathology AI Hackathon, CODiPAI Challenge Oct. 2023
MENTORING	[a.1] SNUAIMED Excellence Scholarship 2023, 2024
	[t.4] Teaching Assistant, Korea Clinical Datathon (KCD) 2025 Oct. 2025
	[t.3] Student Ambassador, MICCAI 2025 AMAI Workshop Sep. 2025
	[t.2] Teaching Assistant, Clinical and Healthcare AI Training Programs 2024, 2025
	[t.1] Teaching Assistant, NVIDIA MONAI Bootcamp, HCLS Summit Korea 2022 May 2022
PATENTS	[p.2] Y.-G. Kim, . . . , B. Jang . Controllable Normal-to-Disease Fundus Translation via Factorial Guidance. Korea Patent Application No. 10-2026-0119988, filed June 2026. (<i>Pending</i>)
	[p.1] H.-S. Kim, . . . , Y.-G. Kim, B. Jang . Method and System for Predicting Information Related to Abdominal Surgery from Medical Image Data. Korea Patent No. 10-2953173, registered April 2026. (<i>Granted</i>)
REFERENCES	Prof. Jin-Wook Choi, PhD (jinchoi@snu.ac.kr) Prof. Young-Gon Kim, PhD (younggon2.kim@gmail.com) Prof. Hyuk Jin Choi, MD, PhD (docchoi@hanmail.net)